

Arithmetic expression outside of simple scalar multiplication is currently not supported

<https://github.com/tensorflow/quantum/issues/565>

ROW 33

Arithmetic expression outside of simple scalar multiplication is currently not supported #565

[New issue](#)[Closed](#)

theRoughCode opened on May 7, 2021 · edited by theRoughCode

Edits · ...

I'm trying to simulate the bias and weights in a classical neuron using `sympy` as follows:

```
qubits = cirq.GridQubit.rect(1, 1)
out_bits = qubits[-1:]
readouts = [cirq.Z(bit) for bit in out_bits]

encoding_params = sympy.symbols('enc0:{}'.format(self.input_dims))
weights = sympy.symbols('weights0:{}'.format(self.num_weights))
bias = sympy.symbols('bias0:{}'.format(self.num_bias))
params = [b + w * x for x, w, b in zip(encoding_params, weights[:3], bias[:3])]

circuit = cirq.Circuit(
    cirq.rx(params[0])(qubits[0]),
    cirq.ry(params[1])(qubits[0]),
    cirq.rz(params[2])(qubits[0])
)

pqc = tfq.layers.ControlledPQC(circuit, readouts)
```

However, I'm getting the following error: `ValueError: Arithmetic expression outside of simple scalar multiplication is currently not supported. See serializer.py for more information.`

It seems like this sort of arithmetic manipulation of symbols is not supported yet. Is there a way around this?



Assignees

No one assigned

Labels

No labels

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Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

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Participants



lockwo on May 8, 2021 · edited by lockwo

Edits · Contributor · ...

I believe that comes from the fact that your params looks like `[bias0 + enc0*weights0, bias1 + enc1*weights1, bias2 + enc2*weights2]` and only multiplication by a constant is allowed e.g. `[10*weights0, 2*weights1, 4*weights2]`. You could probably get around this with a custom layer. This seems to be similar to what you are trying to achieve:

```
class bias_layer(tf.keras.layers.Layer):
    def __init__(self, num_params, readout_ops):
        super(bias_layer, self).__init__()
        self.w = tf.Variable(initial_value=np.random.uniform(0, 2 * np.pi, (1, num_params)), dtype="float32",
                             self.b = tf.Variable(initial_value=np.random.uniform(0, 2 * np.pi, (1, num_params)), dtype="float32",
                             self.layer = tfq.layers.ControlledPQC(circuit, readout_ops)

    def call(self, inputs):
        return self.layer([inputs, self.w + self.b])
```



MichaelBroughton on May 10, 2021

Collaborator · ...

Thanks for jumping on this @lockwo! Building on that point, we only support scalar multiplication in circuit symbols and not much else. If you want to Compute any kind of complicated expression involving your symbol before putting it into a circuit, you should do so in TF. i.e.:

```
my_variable = <some tf.Tensor>
step1 = tf.math.log(my_variable)
step2 = step1 ** 2 + 10
step3 = step2 / 3.14
```



Zotero



MichaelBroughton on May 10, 2021

Collaborator ...

Thanks for jumping on this @lockwo ! Building on that point, we only support scalar multiplication in circuit symbols and not much else. If you want to Compute any kind of complicated expression involving your symbol before putting it into a circuit, you should do so in TF. i.e.:

```
my_variable = <some tf.Tensor>
step1 = tf.math.log(my_variable)
step2 = step1 ** 2 + 10
step3 = step2 / 3.14
other_values = <some other tf.Tensor>
step4 = tf.math.cos(other_values)

final_inputs = tf.concat([step3, step4], ...)
a_pqc = tfq.layers.ControlledPQC(...)([..., final_inputs])
```



The above will feed in values to the symbols in your circuit that have had those transformations applied beforehand. We didn't want to find ourselves re-implementing all this basic math stuff on our end when TF can already do it very reliably and quickly for us.

Does this help clear things up ?



MichaelBroughton on Jun 8, 2021

Collaborator ...

Going to close for now. Feel free to re-open if there's any more issues.



MichaelBroughton closed this as completed on Jun 8, 2021



theRoughCode on Jul 3, 2021

Author ...

That definitely helps, sorry for the late response!



lockwo mentioned this on Jun 21, 2022

🔗 Arithmetic expression outside of simple scalar multiplication is currently not supported. #693

Arithmetic expression outside of simple scalar multiplication is currently not supported. #693

New issue

Closed



Gopal-Dahale opened on Jun 21, 2022

...

I created a parameterized circuit using Cirq for Amplitude Encoding as described in [Supervised Learning with Quantum Computers](#). The parameter is an arithmetic expression involving multiple operations like division and squaring. Below is the formula used. This value is used inside the R_y gate.

$$\beta_j^s = 2\arcsin\left(\frac{\sqrt{\sum_{l=1}^{2^g-1} |A_{(2j-1)2^g-l+1}|^2}}{\sqrt{\sum_{l=1}^{2^g} |A_{(j-1)2^g+l}|^2}}\right).$$

Using the circuit in the `tfq.layers.ControlledPQC` gives the following error.

ValueError: Arithmetic expression outside of simple scalar multiplication is currently not supported. See `serializer.py` for more information.



Any hack for this?



lockwo on Jun 21, 2022

Contributor ...

Hard to say the best solution without seeing the exact code causing you trouble. But based on the error I would guess you are doing some disallowed symbol operations and that it could probably be done in a custom layer. I recommend [#565](#) which had the same problem/error



lockwo on Aug 15, 2022

Contributor ...

Any updates on this or should it be closed?



1



mhucka on Dec 1, 2024

Member ...

The OP appears to have given the thumbs-up to closing this issue, so I'm going ahead and closing it.



mhucka closed this as **completed** on Dec 1, 2024

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